13.5 Weighted Averages

Lesson Introduction

Benchmark	MA.912.GR.3.1 Determine the weighted average of two or more points on a line.	Learning Targets	• I can determine the weighted average of two or more points on a line. • I can explain how changing the weight moves the weighted average of points.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• How can you determine the weighted average of two or more points on a line? • How can you explain how changing the weight moves the weighted average of points?
Lesson Narrative	This lesson applies knowledge from Lesson 13.4, to guide students through explorations on finding the weighted average of two or more points on a line. The lesson relates partitioning line segments to the definition of a weighted average, and how to find the weighted average on the coordinate plane using a given ratio.

Component Summary	13.5.1 Warm-Up (~5 mins.)	Stock market and growth of shares *(SE p. 306)*	13.5.4 Collaboration (5 mins.)	Using weighted averages to dilate polygons *(SE p. 312)*
	13.5.2 Exploration (10 mins.)	Discovery on weighted averages *(SE p. 307)*	13.5.5 Your Turn! (5 mins.)	Independent practice problems on weighted averages *(SE p. 313)*
	13.5.3 Guided Instruction (20 mins.)	Weighted averages on the coordinate plane *(SE p. 309)*	13.5.6 Wrap-Up (~5 mins.)	Ticket out the door *(SE p. 314)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • N/A <u>Additional Terms:</u> • Dilation	• (Optional): Fishing Weight • (Optional): Pencil to tie fishing weight to • (Optional): String to tie the fishing weight to pencil	• If choosing to do so, prepare the string, fishing weight, and pencil.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>On 13.5.3 Guided Instruction, students may struggle with writing expressions using given ratios with weighted averages.
Students may struggle with visualizing weighted averages on the graph in 13.5.3 Guided Instruction.
Students may struggle with dilating using segment partitioning in 13.5.4 Collaboration.</td><td>Consider using a fishing weight and a pencil to demonstrate the idea of weighted averages in 13.5.3 Guided Instruction.
13.5.4 Collaboration can be used as enrichment.
If time is an issue, then consider assigning 13.5.5 Your Turn! for homework.
Consider using the Desmos graph that is linked for question 5 in 13.5.3 Guided Instruction to move the point to help students further relate the weighted average to its graphical representation.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Three Reads
The concept of weighted averages on a line can confuse students at first. To help guide students through any misconceptions, implement a Three Reads on 13.5.3 Guided Instruction. Allow students to read through the scenario one time, silently. Have students read again and highlight any key points or concepts in the second read. On the third read, students should read again, either silently or aloud. Then, have students summarize what they read with a partner.</td><td>MA.K12.MTR.2.1: Multiple Representations
Lesson 13.5 requires that students view the problems from both a graph and an equation. Allow students to make connections between the graphical representations as well as the algebraic representations. Being able to successfully master this standard will require students to progress from modeling with graphs to using expressions.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider using a manipulative for modeling the concept of a weighted average. A ruler or pencil with a fishing weight can be used to model the concept of a weighted average. For a digital approach, consider finding a video or applet that models the concept of a weighted average.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

13.5.1 Warm-Up: Between Two Numbers

Component Overview: *Students begin by reading a scenario about the stock market and answer two questions.*

13.5.1 Warm-Up: Between Two Numbers

A stock exchange, or stock market, is a system for buying and selling bonds, securities, or stocks. A stock is a share in the ownership of a company. NASDAQ is short for National Association of Securities Dealers Automated Quotations, which is the world's largest electronic stock exchange. Each company that is traded has an abbreviation of one to four letters.

Jaxon's grandmother bought 100 shares of Netflix and 100 shares of Apple for him when he was born on March 29, 2003. She paid \$1.37 per share for Netflix (NASDAQ: NFLX) and paid \$0.21 per share for Apple (NASDAQ: AAPL).

On Jaxon's 18th birthday, Netflix's stock price was at \$513.95 and Apple's stock price was at \$121.39.

1. How much were the stocks worth on Jaxon's 18th birthday?
   \$63,534.00

The stock price history of Netflix shows that at one point the stock had increased in worth by 45,000%, and Apple stock had increased in worth by 140,000%.

2. Write the decimal equivalent of each percentage increase.

Stock	Decimal Equivalent
Apple	1400
Netflix	450

The stock market is a real-world context for mathematics that can effectively garner student interest, with a variety of opportunities for meaningful connections to mathematics in the real world (MTR 7).

Consider one of the following strategies for embedding this lesson in real-world contexts:

- *Show a video on the stock market, or popular mainstream applications of stocks.*
- *Have students participate in stock market simulations as an extension project.*
- *Partner with a local company to provide students with real-life stocks to monitor. For example, some apps give free stock just for signing up.*

13.5.2 Exploration: Exploring Weighted Averages

Component Overview: *Students explore weighted averages by starting with midpoints (ratio 1:1); then, students discover the mathematics behind weighted averages using a graphical representation.*

13.5.2 Exploration: Exploring Weighted Averages

1. A number line is shown. Each grid space is 1 unit.

a. Find the midpoint of $\overline{HM}$ and label it B.

Assuming that points H and M have the same weight, the diagram represents how points H, B, and M are in balance.

b. Discuss with your partner why point B acts as a point of balance. Summarize your discussion.
Answers will vary. Point B is the midpoint of line segment HM so the distance from H to B is the same as the distance from B to M so point B will balance H and M.

c. Keyshawn told his partner that if the scale is drawn where the scale is centered on point K, then point H would be higher than point M. Describe or draw what would have to be done to balance the scale when it is centered at point K.
Answers will vary. The side of the scale with point H will need to be lowered. To lower it another point H could be added.

The ratio of $HK:KM$ is $3:9$, which can be written as $\frac{1}{3}$. The ratio, $\frac{1}{3}$, can be used to determine the location of point K using the steps shown.

$$\frac{3}{4}(\text{point } H) + \frac{1}{4}(\text{point } M) = \frac{3}{4}(3) + \frac{1}{4}(15) = \frac{9}{4} + \frac{15}{4} = \frac{24}{4} = 6$$

This Exploration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the 13.5.3 Guided Instruction. Students may revise their thinking throughout the Exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group 13.5.3 Guided Instruction.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

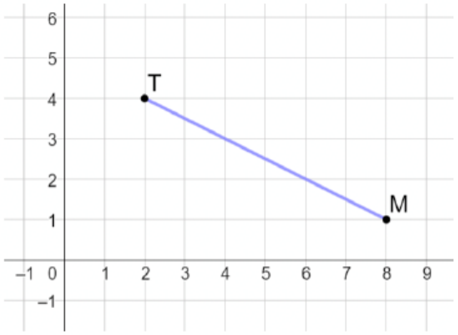
Students may struggle with answering question 1c. Demonstrating a scale, see-saw, or balance may help students visualize that more weight should be added. Students may not connect what that the weight should be H.

- *How do the fractions, $\frac{3}{4}$ and $\frac{1}{4}$, relate to the ratio $\frac{1}{3}$?*
- *How does the coefficient of H indicate that point H is weighted?*

13.5.2 Exploration: Exploring Weighted Averages (continued)

The ratio $\frac{1}{3}$ was used to find $\frac{3}{4}$ and $\frac{1}{4}$ by noting that the ratio indicates there are a total of 4 parts. To balance at point K, point H would be "heavier" than point M, so it is weighted as 3 of the 4 parts with point M weighted as 1 of the 4 parts.

2. Point T has coordinates $(2, 4)$, and point M has coordinates $(8, 1)$.



a. What is the midpoint of line segment TM?

$\left(\frac{2+8}{2}, \frac{4+1}{2}\right) = (5, 2.5)$

b. Complete the table.

	x-value	y-value
$T(2, 4)$	$\frac{1}{2}(2)$	$\frac{1}{2}(4)$
$M(8, 1)$	$\frac{1}{2}(8)$	$\frac{1}{2}(1)$
Total	5	2.5

c. Discuss with your partner how the table shows the process of finding the midpoint.

Answers will vary. Find half of each point's coordinate value and then add them together.

- *Can you use the graph to find the midpoint? If so, how?*
- *What do the ½'s represent in the table?*

Consider allowing students times to write down things they notice and wonder about the table before discussing with a partner.

d. Complete the expression that can be used to find the point that partitions the line segment in a $1:1$ ratio. T represents the coordinates of point T, and M represents the coordinates of point M.

$$\frac{1}{2}T + \frac{1}{2}M$$

e. Find the point that partitions directed line segment TM in a $2:1$ ratio.
$(6, 2)$

f. Complete the table to calculate $\frac{1}{3}T + \frac{2}{3}M$.

	x-value	y-value
$T(2, 4)$	$\frac{1}{3}(2)$	$\frac{1}{3}(4)$
$M(8, 1)$	$\frac{2}{3}(8)$	$\frac{2}{3}(1)$
Total	6	2

g. What do you notice about the total in the table and the point found in part e that partitions the directed line segment in a $2:1$ ratio?
They are the same.

3. Line segment KR has endpoints at $K(-6,1)$ and $R(2,5)$.

a. Complete the expression for the point that partitions directed line segment KR in a $3:1$ ratio.

$$\frac{1}{4}K + \frac{3}{4}R$$

b. Use the expression to find the point that partitions directed line segment KR in a $3:1$ ratio.

$\frac{1}{4}(-6) + \frac{3}{4}(2) = 0$

$\frac{1}{4}(1) + \frac{3}{4}(5) = 4$

$(0,4)$

- *Why does the ratio of 1:1 create the equation of* $\frac{1}{2}T + \frac{1}{2}M$*?*
- *Which strategy works better for finding the point that partitions a directed line segment, the table or a formula? Explain your choice.*

As you monitor students, listen for misconceptions or confusion on why a ratio of 2:1 relates to the formula. Consider how to support student learning during 13.5.3 Guided Instruction. Students may benefit from reviewing the questions 2f, 2g, and 3 at the start of Guided Instruction.

13.5.3 Guided Instruction: Determining Weighted Averages

Component Overview: *Students are introduced to the concept of weighted averages on a line. Students are guided through a scenario about a seesaw, and how weight distribution on a see-saw works.*

13.5.3 Guided Instruction: Determining Weighted Averages

Weight is a measure of force pushing or pulling on an object due to gravity. This means that finding a weighted average puts more "weight" on one point than another.

If B is weighted twice as heavily as A, then during the calculation, B is to measure two times the value of A. In other words, B has two times the "gravitational pull" of A.

Think of it in terms of a seesaw. In the picture below, point B is weighted more heavily than point A.

1. Discuss with your partner what the girls close to point B would have to do so that the seesaw would balance. Summarize your discussion.
   To balance the seesaw, the girls on the heavier end would need to shift closer to the middle. This can also be shown by shifting the balancing point closer to the heavier object.

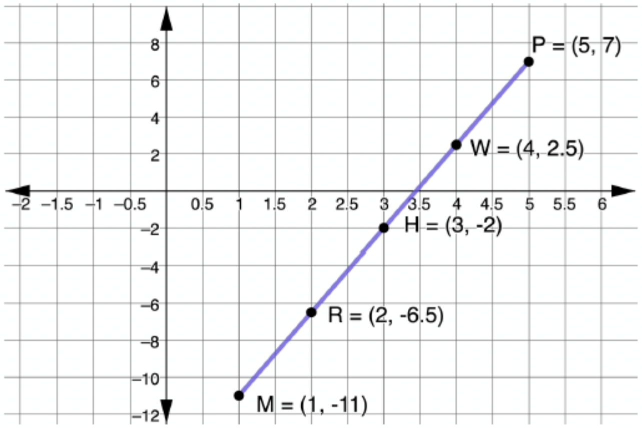
Have students read this section 3 times. See the Language and Literacy Routine on page 2 for further guidance.

- *What do you notice about the picture?*
- *How does this picture relate to the lesson and standard?*

It may be helpful to demonstrate the balance of an object where one end is heavier than another. As in the seesaw picture, to balance the seesaw, the girls on the heavier end would have to shift closer to the middle. This can also be shown by shifting the balancing point closer to the heavier object. Consider using a fishing weight and a pencil to demonstrate this idea. See "Differentiate Instruction" for other ideas.

13.5.3 Guided Instruction: Determining Weighted Averages (continued)

Directed line segment MP is shown, where point M is at $(1, -11)$ and point P is at $(5, 7)$. Point H is the midpoint of $\overline{MP}$, point W is the midpoint of $\overline{HP}$, and point R is the midpoint of $\overline{MH}$.



Since point H is the midpoint of $\overline{MP}$, the coordinates of point H can be determined by finding the average of points M and P.

$$\frac{M+P}{2}$$

Point W partitions the line segment MP so that point P weighs three times as much as point M. This means that point W is closer to point P because point P is "heavier" than point M. The given expression can be used to calculate this weighted average when point P weighs three times as much as point M.

$$\frac{M+P+P+P}{4}$$

Have students either underline or highlight the information at the top of this page. Consider doing another Three Reads on this page as well. See guidance on Language and Literacy Routine.

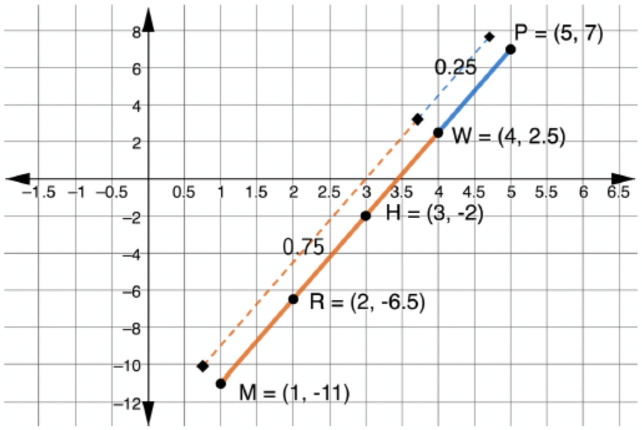
Discuss with students how the expression can also be written as $\frac{1}{2}M + \frac{1}{2}P$, and $\frac{1}{4}M + \frac{3}{4}P$.

Consider asking the following questions to help guide students through this page:

- *Why is the equation $\frac{M+P}{2}$?*
- *How was this equation derived?*
- *At the bottom of the page, why is the equation $\frac{M+P+P+P}{4}$?*
- *How was this equation derived?*

13.5.3 Guided Instruction: Determining Weighted Averages (continued)

The graph shows the weighted averages.



2. Discuss with your partner why the distance from point W to point P is shorter if point P is "heavier" than point M. Summarize your discussion.
   As in the seesaw picture, to balance the seesaw the girls on the heavier end would need to shift closer to the middle. This can also be shown by shifting the balancing point closer to the heavier object. Point W is the balancing point so would need to be closer to point P because point P is "heavier."

3. Point R partitions the line segment MP so that point M weighs three times as much as point P. Write an expression that shows this weighted average.
   $\frac{M+M+M+P}{4}$

A midpoint calculation weighs points on a line segment equally. In a **weighted average** calculation, points on a line segment can be weighted differently.

Students may struggle with conceptualizing the graph on this page. Consider providing students with additional visualization and explanation, if necessary.

Have a couple of students share their partner discussions with the class to get a variety of perspectives on this question.

How can you write an expression that models point M having three times as much weight as point P?

Consider revisiting question 1 in the Collaboration to make the connection to the balance questions.

13.5.3 Guided Instruction: Determining Weighted Averages (continued)

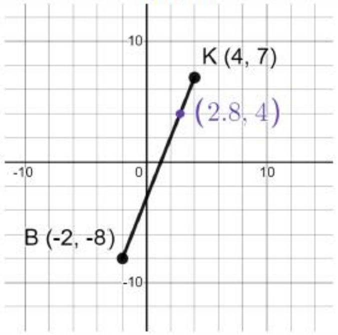
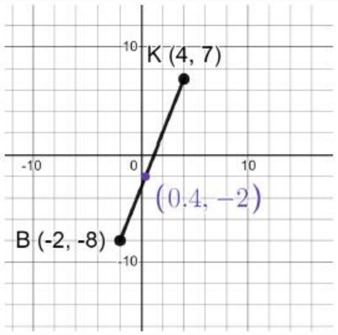
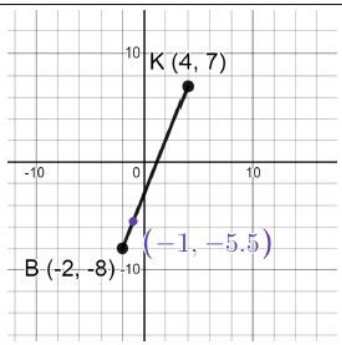
4. Write the expression that can be used to find the point that partitions $\overline{MP}$ in the given ratio, and find the endpoint that is the weighted point.

Ratio	Expression	Point	Weighted Point
1: 1	$\frac{M+P}{2}$	$\frac{1+5}{2}, \frac{-11+7}{2} = (3, -2)$	*Neither*
1: 3	$\frac{M+M+M+P}{4}$	$\frac{1+1+1+5}{4}, \frac{-11-11-11+7}{4} = (2, -6.5)$	M
3: 1	$\frac{M+P+P+P}{4}$	$\frac{1+5+5+5}{4}, \frac{-11+7+7+7}{4} = (4, 2.5)$	P

Students may struggle with writing the expression in the table on question 4. If necessary, write the expression and have students explain it. Emphasize the reason that a point is repeated is because it is the weighted point.

13.5.3 Guided Instruction: Determining Weighted Averages (continued)

5. A line segment is shown in the table with three different weighted averages. Write the expression that can be used to find each weighted average.

Partition	Graph	Expression
$4:1$		$\frac{B+K+K+K+K}{5}$ *or* $\frac{1}{5}B+\frac{4}{5}K$
$2:3$		$\frac{B+B+B+K+K}{5}$ *or* $\frac{3}{5}B+\frac{2}{5}K$
$1:5$		$\frac{B+B+B+B+B+K}{6}$ *or* $\frac{5}{6}B+\frac{1}{6}K$

Use the Desmos graph embedded in the digital Student Edition to show students the distances and ratios for the graphs in question 5 by moving the purple point along the line.

Remind students to write the expression by using the weighted point. Avoid providing students with a quick summary such as "use the second number of the ratio as the numerator of the first fraction, and the first number of the ratio as the numerator of the second fraction". Students may find such a summary difficult to remember. Focus on students understanding why that summary works.

- *Why is B used as the first letter in the expression?*
- *Why is K used as the second letter in the expression?*
- *What is your thought process when writing the expression?*

13.5.3 Guided Instruction: Determining Weighted Averages (continued)

6. Discuss with your partner how changing the weights moves the weighted average on the number line. Summarize your discussion.
As in the seesaw picture, to balance the seesaw the girls on the heavier end would have to shift closer to the middle. This can also be shown by shifting the balancing point closer to the heavier object. The weighted average shifts closer to whichever endpoint is "heavier" according to the ratio.

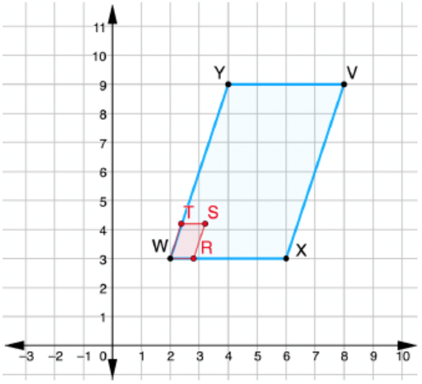
Allow students to share their discussions with the class to hear a variety of student perspectives.

13.5.4 Collaboration: Weighted Averages and Dilations

Component Overview: *Students find the coordinates of the image of given polygons by using weighted averages and segment partitioning.*

13.5.4 Collaboration: Weighted Averages and Dilations

1. $VXWY$ has vertices at $V(8,9)$, $X(6,3)$, $W(2,3)$, and $Y(4,9)$.



a. Find the point that partitions $\overline{WX}$ in a $1:4$ ratio. Label it R.

$\frac{4}{5}(2)+\frac{1}{5}(6)=2.8$

$\frac{4}{5}(3)+\frac{1}{5}(3)=3$

$(2.8,3)$

b. Find the point that partitions $\overline{WY}$ in a $1:4$ ratio. Label it T.

$\frac{4}{5}(2)+\frac{1}{5}(4)=2.4$; $\frac{4}{5}(3)+\frac{1}{5}(9)=4.2$

$(2.4,4.2)$

c. Find the point that partitions $\overline{WV}$ in a $1:4$ ratio. Label it S.

$\frac{4}{5}(2)+\frac{1}{5}(8)=3.2$; $\frac{4}{5}(3)+\frac{1}{5}(9)=4.2$

$(3.2,4.2)$

d. Is $WTSR$ a dilation of $WYVX$? Justify your answer.

Yes, it is a dilation of $\frac{1}{5}$ about the point W.

Consider using this component for enrichment. Consider using Stepping Stones in place of this component for students who struggle with weighted averages.

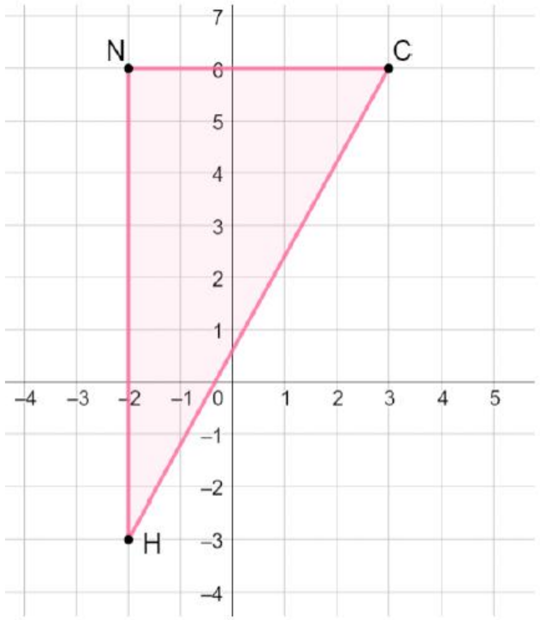
Alternatively, this component can be completed by assigning different students parts a, b, and c. Then, pair students with a different part to have each partner explain how they determine the answer. Then, have each pair do the part that neither was assigned.

How can an equation be set up to find the partitioned points?

- *How can you determine whether $WTSR$ is a dilation of $WYVX$?*
- *How can you find the scale factor of the dilation?*

13.5.4 Collaboration: Weighted Averages and Dilations (continued)

2. Triangle NCH has vertices at $N(-2,6)$, $C(3,6)$, and $H(-2,-3)$.



Use segment partitioning to dilate ΔNCH using center C and scale factor $\frac{3}{4}$.

N': $\frac{3}{4}N + \frac{1}{4}C \rightarrow (-0.75, 6)$

H': $\frac{3}{4}H + \frac{1}{4}C \rightarrow (-0.75, -0.75)$

Students may struggle with using segment partitioning to find the scale factor. Allow students to use the graph to plot points. Consider using the guiding questions below to help students.

If using this component for enrichment provide students the opportunity to explore dilations using online geometry tools such as GeoGebra. Pair students together and have each create a triangle and apply a dilation using a scale factor of their choice or of your choosing. The center of dilation should be a vertex. Each pair writes down the coordinates of the original triangle and of the dilation and switches the information with another pair. The other pair uses the coordinates to determine the center of dilation and the scale factor. Students may choose to do this by trial and error or algebraically using $\frac{a}{b}$ to represent the scale factor. For example, from question 2 the equation $\frac{a}{b}(-2) + \left(1 - \frac{a}{b}\right)(3) = -0.75$ can be used to determine the scale factor $\frac{a}{b}$.

Use the following questions to guide students through dilating ΔNCH, using center C and a scale factor of ¾

- *How long is line segment NC? (5)*
- *What is ¾ of 5? (3.75)*
- *What is the coordinate that is 3.75 units from point C? (-0.75,6)*
- *How long is line segment NH? (9)*
- *What is ¾ of 9? (6.75)*
- *What is the point that is 6.75 units away from point N'? (-0.75, -0.75)*

13.5.5 Your Turn!

Component Overview: *Students are given two problems to work independently.*

13.5.5 Your Turn!

1. Line segment MK has endpoints at $M(-2,1)$ and $K(8,7)$.
   a. Write an expression for the point that partitions directed line segment MK in a $2:3$ ratio.
   $\frac{3}{5}M + \frac{2}{5}K$

   b. Use the expression to find the point that partitions directed line segment MK in a $2:3$ ratio.
   $\frac{3}{5}(-2) + \frac{2}{5}(8) = 2$

   $\frac{3}{5}(1) + \frac{2}{5}(7) = 3.4$

   $(2, 3.4)$

Students may struggle with writing the expression for the point that partitions the directed line segment for the given ratio. Consider using guided questions:

- *Which point weighs more?*
- *How many total parts does line segment MK have?*

13.5.6 Wrap-Up: Ticket Out the Door

Component Overview: *Students complete a Ticket Out the Door.*

13.5.6 Wrap-Up: Ticket Out the Door

1. A number line is shown. Each grid space is 1 unit.

   a. Determine the point on the number line that satisfies the expression shown. Label the point M.

$$\frac{2}{5}(\text{point } W) + \frac{3}{5}(\text{point } T)$$

   b. What is the ratio $WM:MT$?
   6: 4 or 3: 2

How well students perform on the Ticket Out the Door, coupled with what was observed during instruction, can provide information for differentiating instruction and supporting student learning using small group instruction. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.